

Surveys

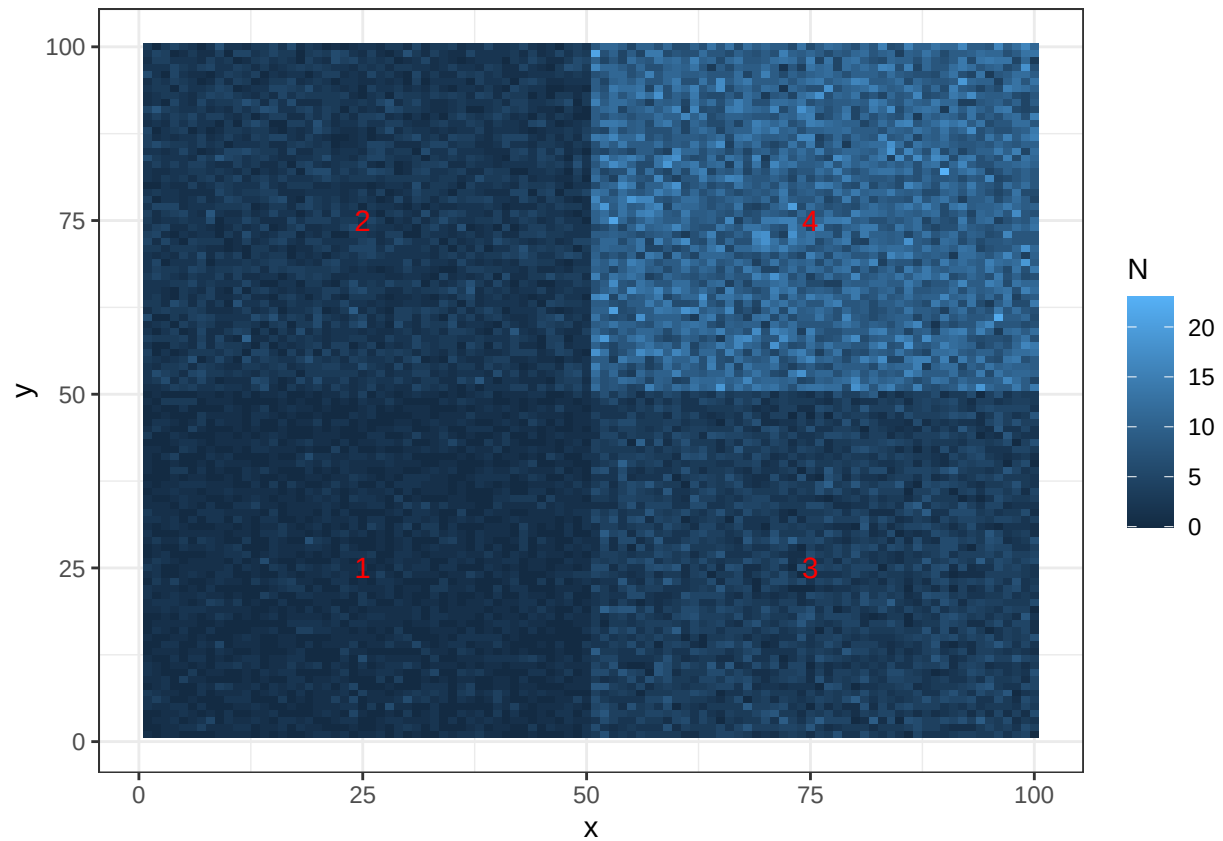
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2025-10-30

```
library(tidyverse)
theme_set(theme_bw())
```

Simulated data for stratification

```
set.seed(10)
meanVals<-c(1,2.4,3.5,10)
area<-expand.grid(x=1:100,y=1:100) %>%
  mutate(strata=case_when(x<=50&y<=50~1,
                           x<=50&y>50~2,
                           x>50&y<=50~3,
                           x>50&y>50~4),
          N=rpois(10000,meanVals[strata])) #random distribution
ggplot(area,aes(x=x,y=y,fill=N))+
  geom_tile()+
  annotate("text",x=c(25,25,75,75),y=c(25,75,25,75),label=1:4,color="red")
```



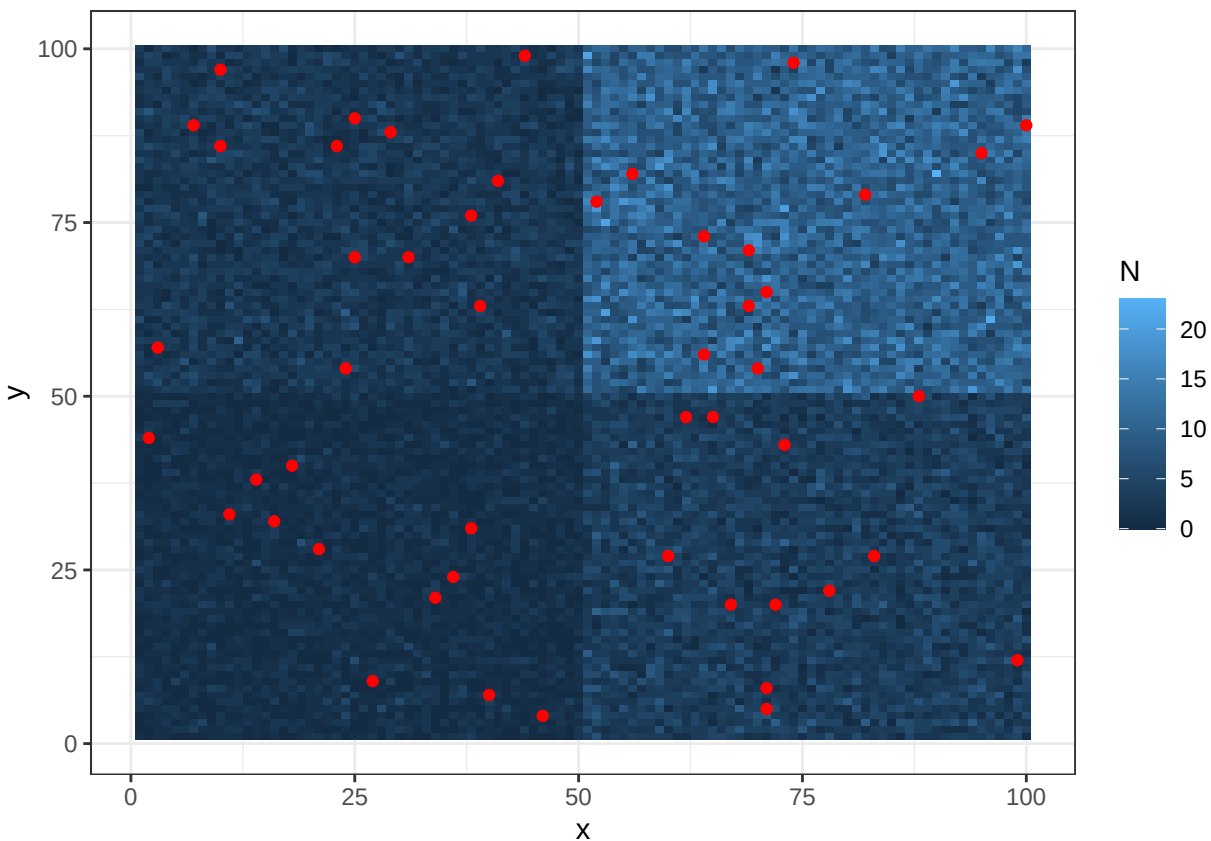
Real population

```
sum(area$N)
```

```
## [1] 42268
```

Make a simple random sample

```
n<-50
dataSimple<-sample(1:nrow(area),size=n,replace=FALSE)
ggplot(area,aes(x=x,y=y,fill=N))+
  geom_tile()+
  geom_point(data=area[dataSimple,],aes(x=x,y=y),color="red")
```



```
simple<-area[dataSimple,] %>%
  summarize(meanCount=mean(N),
            seCount=sd(N)/sqrt(n()),
            n=n(),
            N=meanCount*nrow(area),
            seN=seCount*nrow(area),
            lci=N-2*seN,
            uci=N+2*seN)
simple %>% mutate(across(where(is.numeric),round,2))
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'across(where(is.numeric), round, 2)'.
## Caused by warning:
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
## Supply arguments directly to '.fns' through an anonymous function instead.
##
## # Previously
## across(a:b, mean, na.rm = TRUE)
##
## # Now
## across(a:b, \(x) mean(x, na.rm = TRUE))

## meanCount seCount n      N      seN      lci      uci
## 1         4.1    0.56 50 41000 5644.21 29711.57 52288.43
```

Stratified with same sample size

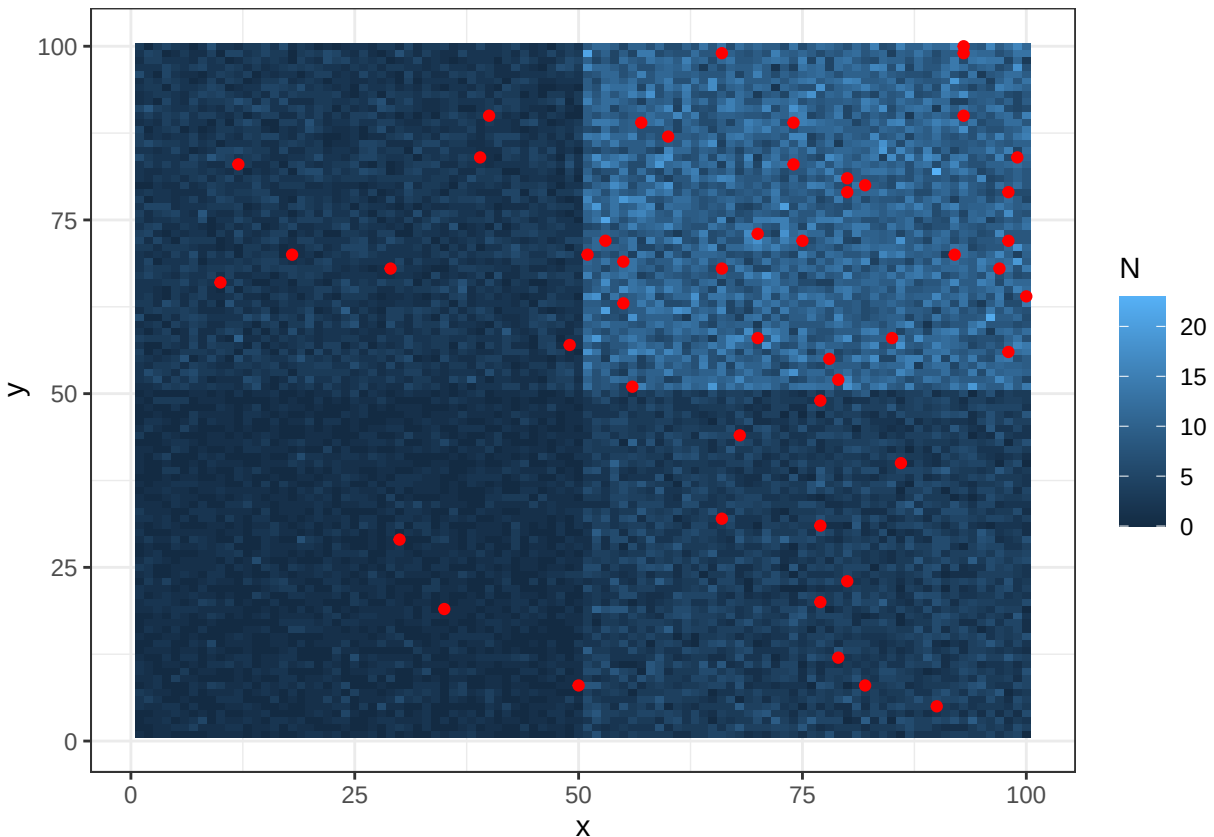
```
SamplesPerStrata<-round(n*meanVals/sum(meanVals)) #variance=mean for Poisson
SamplesPerStrata
```

```
## [1] 3 7 10 30
```

```
strataSample<-NULL
for(i in 1:4) strataSample<-
  c(strataSample,sample(which(area$strata==i),
                        size=SamplesPerStrata[i],
                        replace=FALSE))
table(area$strata[strataSample])
```

```
##
## 1 2 3 4
## 3 7 10 30
```

```
ggplot(area,aes(x=x,y=y,fill=N))+
  geom_tile()+
  geom_point(data=area[strataSample,],aes(x=x,y=y),color="red")
```



```
stratified<- area[strataSample,] %>%
  group_by(strata)%>%
  summarize(meanCount=mean(N),
```

```

      seCount=sd(N)/sqrt(n()),
      n=n(),
      area=nrow(area)/4,
      N=meanCount*area,
      seN=seCount*area) %>%
ungroup()

stratified %>% mutate(across(where(is.numeric),round,2))

```

```

## # A tibble: 4 x 7
##   strata meanCount seCount    n area      N    seN
##   <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
## 1      1      1.33    0.33     3  2500  3333.  833.
## 2      2      2.43    0.43     7  2500  6071. 1071.
## 3      3      2.3     0.45    10  2500  5750. 1121.
## 4      4     10.4    0.58    30  2500 25917. 1446.

```

```

stratTotal<-stratified %>% summarize(N=sum(N),
      seN=sqrt(sum(seN^2)),
      lci=N-2*seN,
      uci=N+2*seN)
stratTotal %>% mutate(across(where(is.numeric),round,2))

```

```

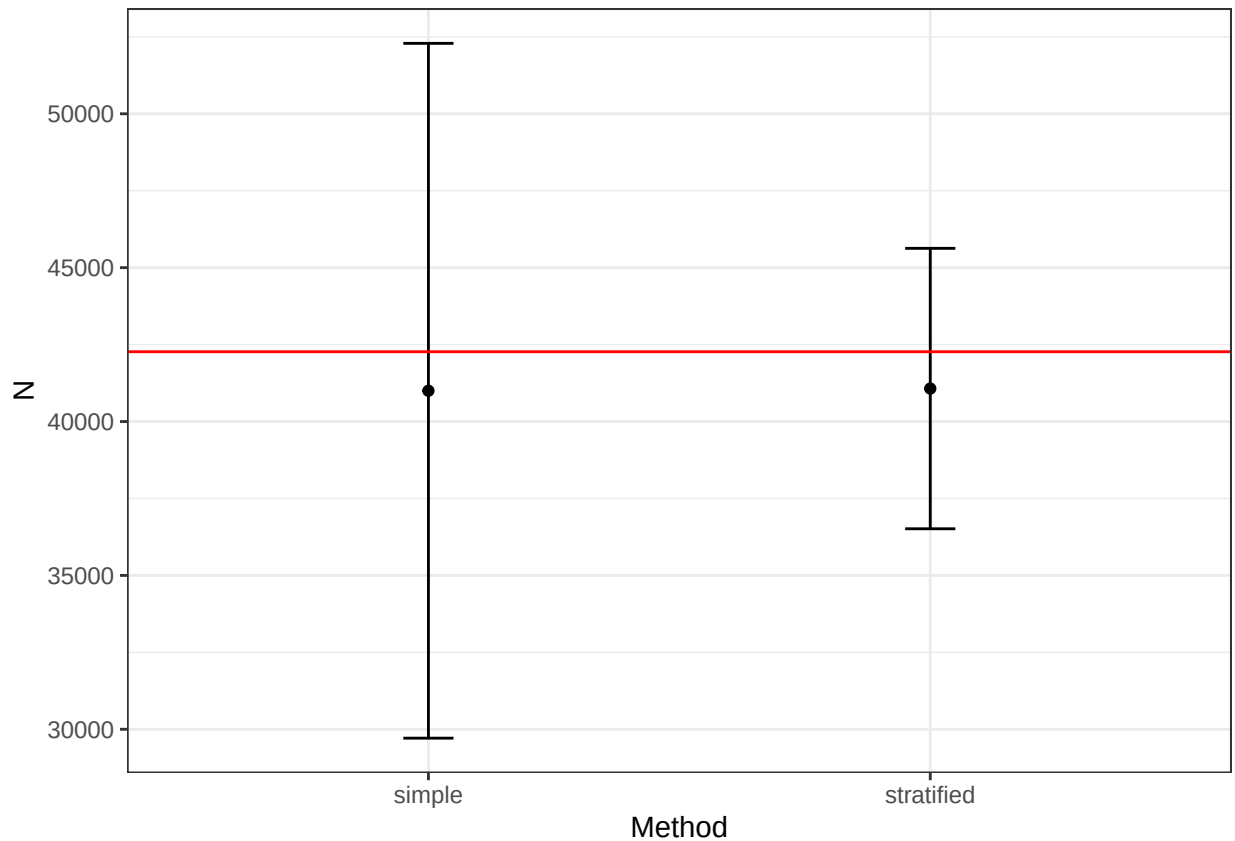
## # A tibble: 1 x 4
##       N    seN    lci    uci
##   <dbl> <dbl> <dbl> <dbl>
## 1 41071. 2278. 36515. 45628.

```

```

bind_rows(list(simple=simple,
      stratified=stratTotal),
      .id="Method")%>%
ggplot(aes(x=Method,y=N,ymin=lci,ymax=uci))+
geom_point()+
geom_errorbar(width=0.1)+
geom_hline(yintercept=sum(area$N),color="red")

```



With finite sample corrections.

```
simpleFinite<-area[dataSimple,] %>%
  summarize(meanCount=mean(N),
            seCount=sd(N)/sqrt(n()),
            f=n/nrow(area),
            n=n(),
            N=meanCount*nrow(area),
            seN=(1-f)*seCount*nrow(area),
            lci=N-2*seN,
            uci=N+2*seN)
simpleFinite %>% mutate(across(where(is.numeric),round,2))
```

```
##   meanCount seCount f  n    N    seN    lci    uci
## 1         4.1   0.56 0 50 41000 5615.99 29768.02 52231.98
```

Makes no difference because sample is a small part of the whole population

Simple random sample and stratified random

Grand Banks Yellowtail Flounder in 2006 from Brad Stevens

```
tows <- read.csv("GB_YTF_2006_Tows2.csv")
head(tows)
```

```
##   Cruise Stratum Tow Num Weight Season area
## 1 200604    1130   1   2   0.30 Spring 0.01
## 2 200604    1130   2   4   1.18 Spring 0.01
## 3 200604    1130   3   6   0.78 Spring 0.01
## 4 200604    1130   4   0   0.00 Spring 0.01
## 5 200604    1130   5   0   0.00 Spring 0.01
## 6 200604    1130   6   5   0.18 Spring 0.01
```

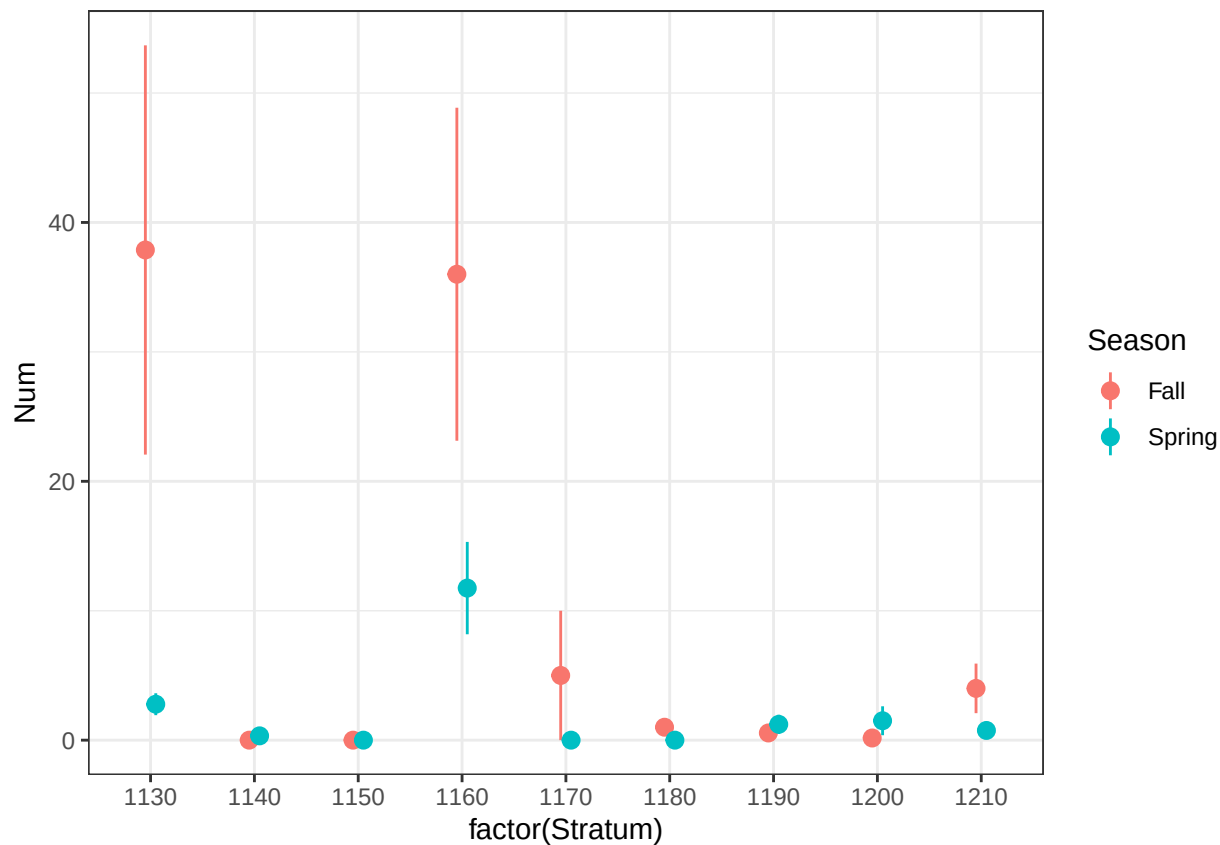
```
strata<-read.csv("strata.csv")
strata
```

```
##   Stratum area
## 1    1130 2374
## 2    1140  656
## 3    1150  230
## 4    1160 2980
## 5    1170  360
## 6    1180  172
## 7    1190 2454
## 8    1200 1221
## 9    1210  424
```

```
ggplot(tows,aes(x=factor(Stratum),y=Num,color=Season))+
  stat_summary(position=position_dodge(width=0.2))
```

```
## No summary function supplied, defaulting to 'mean_se()'
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_segment()').
```



```
with(tows, table(Season, Stratum))
```

```
##           Stratum
## Season   1130 1140 1150 1160 1170 1180 1190 1200 1210
##   Fall      8   3   1   20   3   1   9   6   4
##   Spring     9   3   1   20   3   1   9   6   4
```

Simple random sample in stratum 1160, spring only

```
strataSum<-tows %>%
  filter(Season=="Spring", Stratum==1160)%>%
  group_by(Stratum) %>%
  summarize(meanCount=mean(Num),
            seCount=sd(Num)/sqrt(n()),
            n=n(),
            meanDensity=mean(Num/area),
            seDensity=sd(Num/area)/sqrt(n()),
            sampled.area=sum(area))
#Or, as tows are all the same area (0.01)
strataSum<-strataSum %>%
  mutate(meanDensity2=meanCount/0.01,
         seDensity2=seCount/0.01)
# Use total area to get total numbers
strataSum<-strataSum %>%
  mutate(N=meanDensity*2980,
         seN=seDensity*2980)
```


Total average density and total numbers across seasons and spatial stratas. Calculate totals per stratum, use `left_join` to add total areas to each stratum

```
stratSeasonSum<-tows %>% group_by(Stratum,Season) %>%
  summarize(meanCount=mean(Num),
             seCount=sd(Num)/sqrt(n()),
             n=n(),
             meanDensity=mean(Num/area),
             seDensity=sd(Num/area)/sqrt(n()),
             sampled.area=sum(area)) %>%
  left_join(strata,by="Stratum") %>%
  mutate(N=meanDensity*area,
         seN=seDensity*area) %>%
  ungroup()
```

'summarise()' has grouped output by 'Stratum'. You can override using the
'.groups' argument.

```
view(stratSeasonSum)
```

Some strata are understudied, so can't be estimated. For now, fill in 0 for the variances to get the sum in each season

```
TotalSeason<-stratSeasonSum %>%
  mutate(seN=replace_na(seN,0),
         seDensity=replace_na(seDensity,0)) %>%
  group_by(Season)%>%
  summarize(N=sum(N),
            seN=sqrt(sum(seN^2)),
            lciN=N-2*seN,
            uciN=N+2*seN,
            meanDensity=sum(meanDensity*area)/sum(area),
            seDensity=sqrt((sum(seDensity^2*area)/sum(area))))
```

```
TotalSeason
```

```
## # A tibble: 2 x 7
##   Season      N      seN      lciN      uciN meanDensity seDensity
##   <chr>    <dbl>  <dbl>  <dbl>  <dbl>    <dbl>    <dbl>
## 1 Fall  20243008. 5370655. 9501698. 30984319.    1862.    1005.
## 2 Spring 4697694. 1106130. 2485435.  6909953.     432.     198.
```

For comparison, unstratified sample in each season (not really correct)

```
simpleSum<-tows %>%
  group_by(Season)%>%
  summarise(meanCount=mean(Num),
            seCount=sd(Num)/sqrt(n())) %>%
  mutate(N=meanCount*sum(strata$area)/0.01,
         seN=seCount*sum(strata$area)/0.01,
```

```

lciN=N-2*seN,
uciN=N+2*seN)
simpleSum

```

```

## # A tibble: 2 x 7
##   Season meanCount seCount      N      seN    lciN    uciN
##   <chr>      <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 Fall      19.3     5.63 20971147. 6121854. 8727440. 33214854.
## 2 Spring     5.07     1.44 5513150 1565795. 2381560. 8644740.

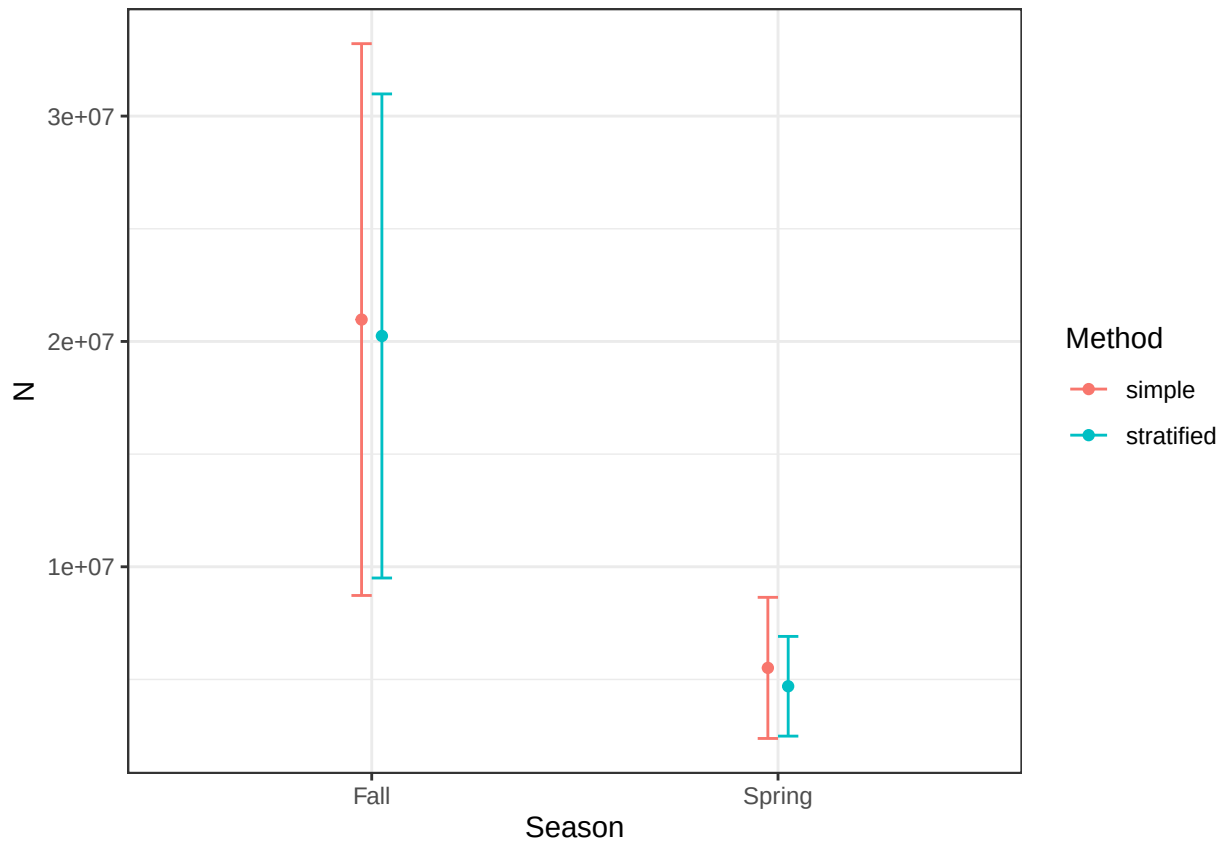
```

Comparing estimates

```

bind_rows(list(simple=simpleSum,
               stratified=TotalSeason),.id="Method") %>%
  ggplot(aes(x=Season,y=N,ymin=lciN,ymax=uciN,color=Method,group=Method))+
  geom_errorbar(position=position_dodge(width=0.1),width=0.1)+
  geom_point(position=position_dodge(width=0.1))

```



In this case it doesn't make much difference if you use simple or stratified

Example 2. Shark bottom longline data

<https://www.fisheries.noaa.gov/inport/item/29395>
snapper-bottom-longline-survey

<https://catalog.data.gov/dataset/shark-and-red->

Strata: 9 – 55 m, 55 – 183 m, and 183 – 366 m, but we will ignore this for now

```
station<-read.csv("Station_Data.csv") %>%
  mutate(date=as.Date(DATE),
         year=year(date))%>%
  filter(year>2000)

catch<-read.csv("Catch_Data.csv")
head(catch)
```

```
##   CDID CID  SID SEAMAPSTATION SAMPLEID      GENUS  SPECIES  BIOCODE SEX
## 1    1    2    1    081523BL04      NA  GALEOCERDO  CUVIER 108022201  F
## 2    2    2    2    081523BL04      NA  GALEOCERDO  CUVIER 108022201  F
## 3    3    2    3    081523BL04      NA  GALEOCERDO  CUVIER 108022201  F
## 4    4    2    4    091423BL01      NA  CARCHARHINUS LIMBATUS 108020205  F
## 5    5    2    5    091423BL01      NA    SPHYRNA  MOKARRAN 108040103  M
## 6    6    2    6    091423BL02      NA  CARCHARHINUS BREVIPINNA 108020207  F
##   PCL SL  FL  TL DW LENGTH WGT INITIALCOND RELCOND FISHID
## 1 1970 NA 2182 2704 NA      M  NA      K      0  AL001
## 2  605 NA  670  938 NA      M 2.6      K      0  AL002
## 3  544 NA  600  842 NA      M 1.6      K      0  AL003
## 4 1100 NA 1230 1560 NA      M  NA      A      1  AL004
## 5 1100 NA 1220 1623 NA      M  NA      A      1  AL005
## 6  740 NA  810 1015 NA      M 4.5      D      0  AL006
##
##           samples TAGNUMBER
## 1 Fin clip,Muscle,Vertebrae
## 2 Fin clip,Muscle,Vertebrae
## 3 Fin clip,Muscle,Vertebrae
## 4           Fin clip  MSU01081
## 5           Fin clip  MSU00686
## 6           Fin clip
##
##                                     COMMENT
## 1 FC-461,many bony catfish fish remains in stomach and crabs,no weight taken
## 2                                     FC-462
## 3                                     FC-463
## 4                                     FC-493,no weight taken
## 5                                     FC-494,no weight taken
## 6                                     FC-495
```

```
view(station)
```

Make a new column for species and count catch by species

```
catchSum<-catch %>%
  mutate(Species=paste(GENUS,SPECIES)) %>%
  filter(Species %in% c("RHIZOPRIONODON TERRAENOVAE","LUTJANUS CAMPECHANUS"))%>%
  group_by(SEAMAPSTATION,CID,Species) %>%
  summarize(count=n())%>%
  pivot_wider(names_from=Species,values_from=count,values_fill = 0)
```

```
## 'summarise()' has grouped output by 'SEAMAPSTATION', 'CID'. You can override
## using the '.groups' argument.
```

```

view(head(catchSum))
StationData<-left_join(station,catchSum,by=c("SEAMAPSTATION","CID"))
StationData<-StationData %>%
  mutate(redSnapper=replace_na(`LUTJANUS CAMPECHANUS`,0),
         AtlSharpnose=replace_na(`RHIZOPRIONODON TERRAENOVAE`,0))
view(StationData)

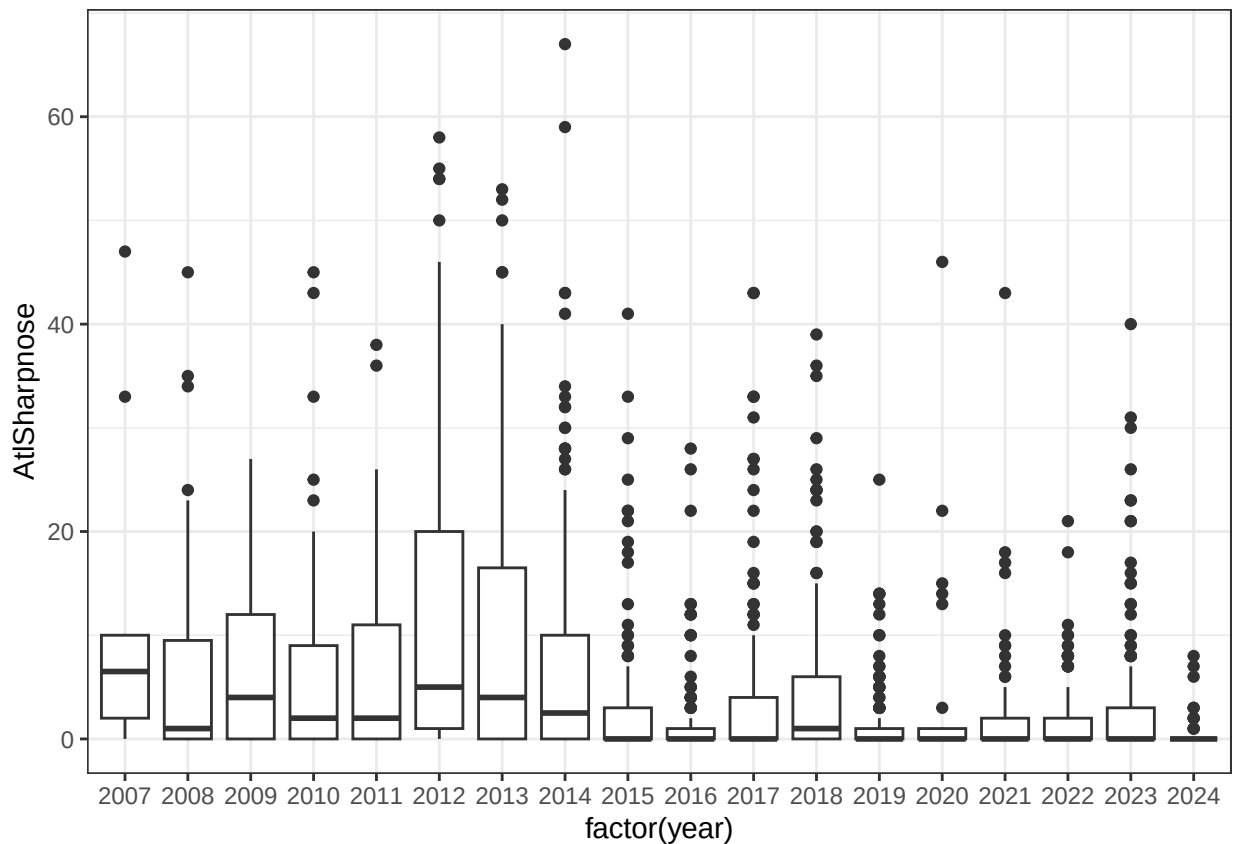
```

Get annual mean numbers

```

ggplot(StationData,aes(x=factor(year),y=AtlSharpnose))+
  geom_boxplot()

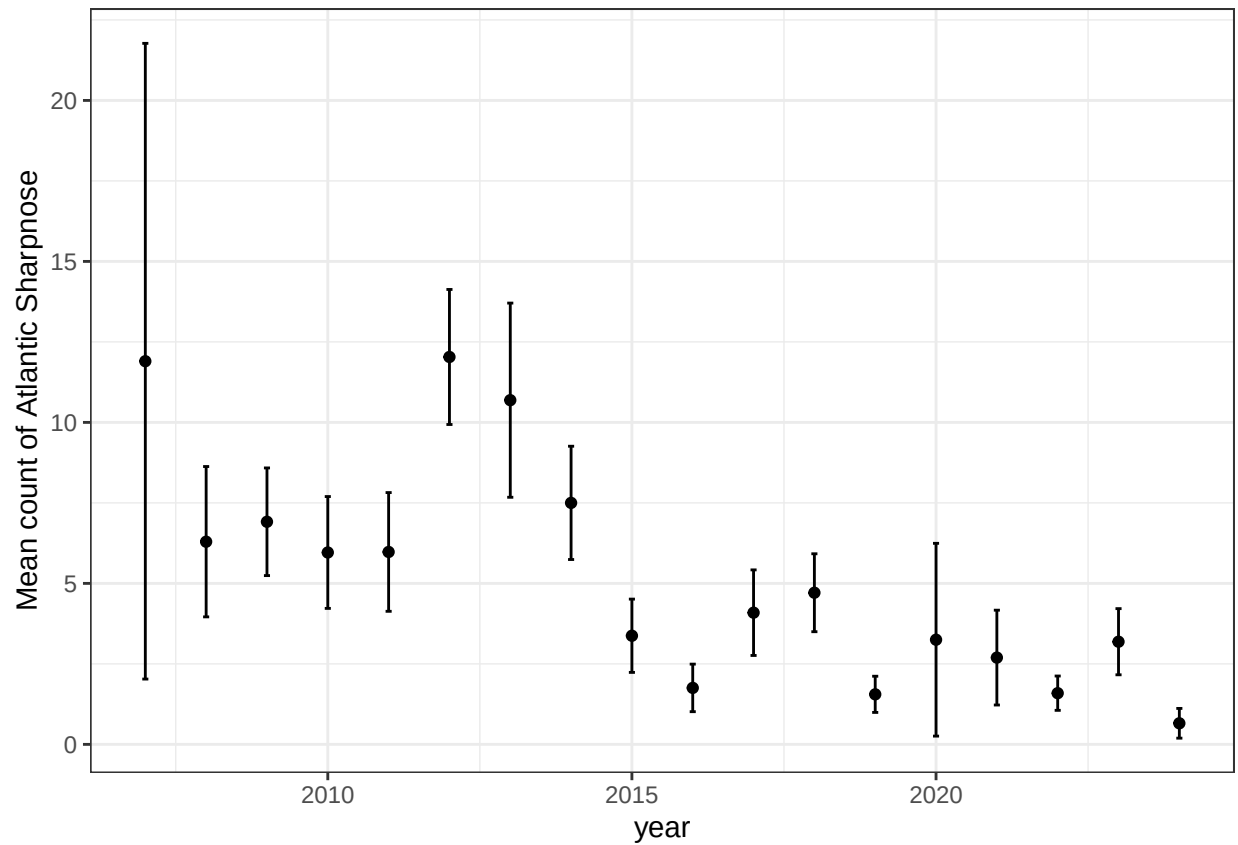
```



```

yearSum<-StationData %>%
  group_by(year) %>%
  summarize(meanAS=mean(AtlSharpnose),
            seAS=sd(AtlSharpnose)/sqrt(n()))
ggplot(yearSum,aes(x=year,y=meanAS,ymin=meanAS-2*seAS,ymax=meanAS+2*seAS))+
  geom_point()+
  geom_errorbar(width=0.1)+
  ylab("Mean count of Atlantic Sharpnose")

```



Distance sampling

<https://distancesampling.org/Distance/articles/lines-distill.html>

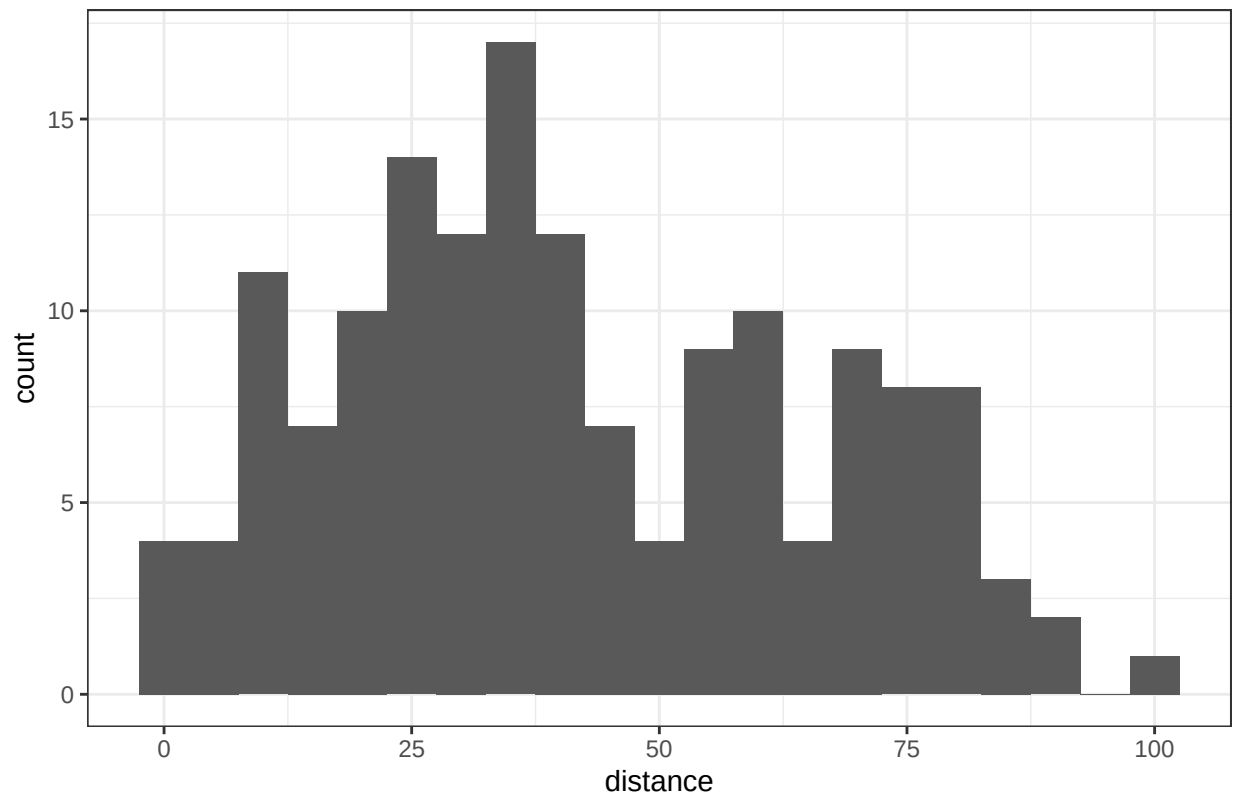
<https://distancesampling.org/Distance/articles/web-only/points/pointtransects-distill.html>

```
library(Distance)
```

Line transect with wrens

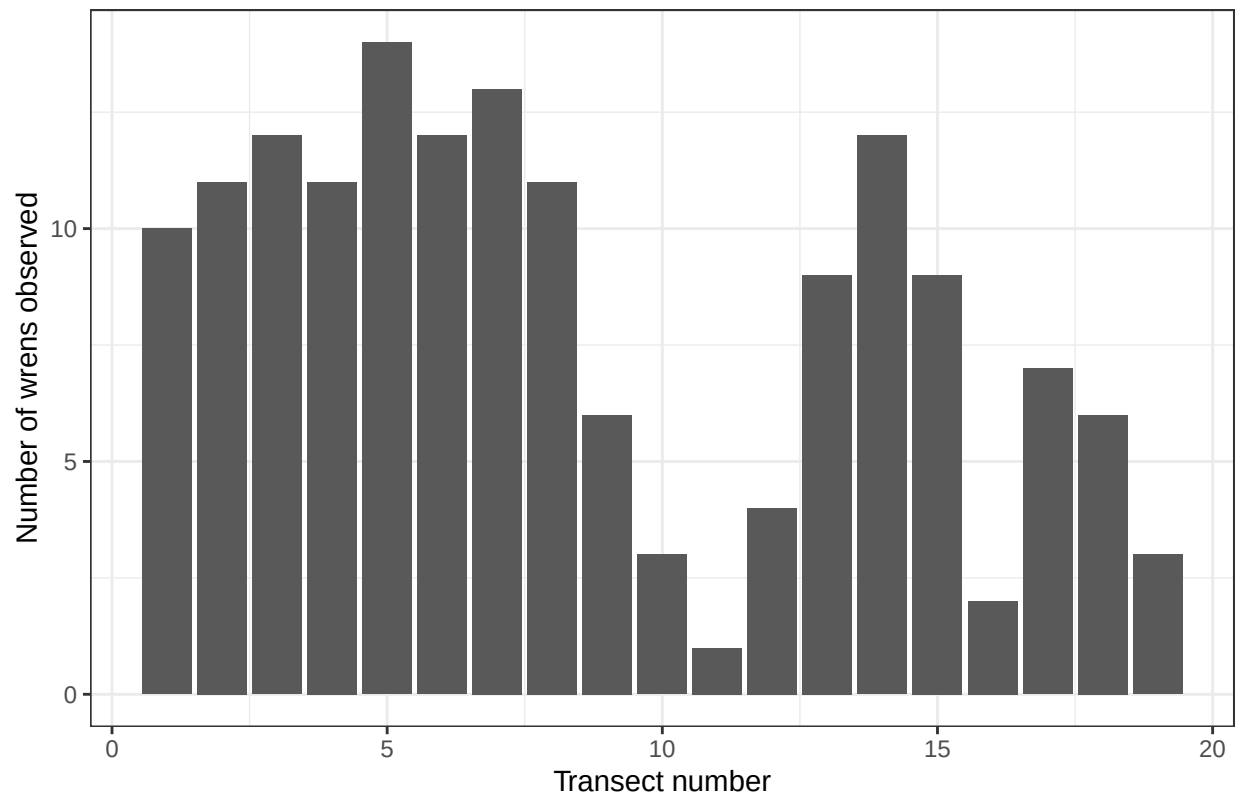
```
data(wren_lt)
ggplot(wren_lt, aes(x=distance)) +
  geom_histogram(binwidth=5) +
  ggtitle("Wren distance from transect line")
```

Wren distance from transect line



```
ggplot(wren_lt,aes(x=as.numeric(Sample.Label)))+  
  geom_bar()+  
  labs(x="Transect number",  
        y="Number of wrens observed",  
        title="Counts of wrens by transect")
```

Counts of wrens by transect



```
head(wren_lt)
```

```
##   Region.Label Area Sample.Label Effort object distance Study.Area
## 1   Montrave 33.2          1 0.416     5      15 Montrave 4
## 2   Montrave 33.2          1 0.416     6      80 Montrave 4
## 3   Montrave 33.2          1 0.416     7      35 Montrave 4
## 4   Montrave 33.2          1 0.416     8      55 Montrave 4
## 5   Montrave 33.2          1 0.416    12      12 Montrave 4
## 6   Montrave 33.2          1 0.416    13      75 Montrave 4
```

Area is the total hectares in the study area. Effort is the length (km) of the transect. Distance is in m. Convert_units function gives units of distance, transect length and area.

Best detection function found with ds()

```
conversion.factor<-convert_units("meter", "kilometer", "hectare")
wren.hn <- ds(data=wren_lt,
              key="hn",
              adjustment=NULL,
              convert_units=conversion.factor)
```

```
## Fitting half-normal key function
```

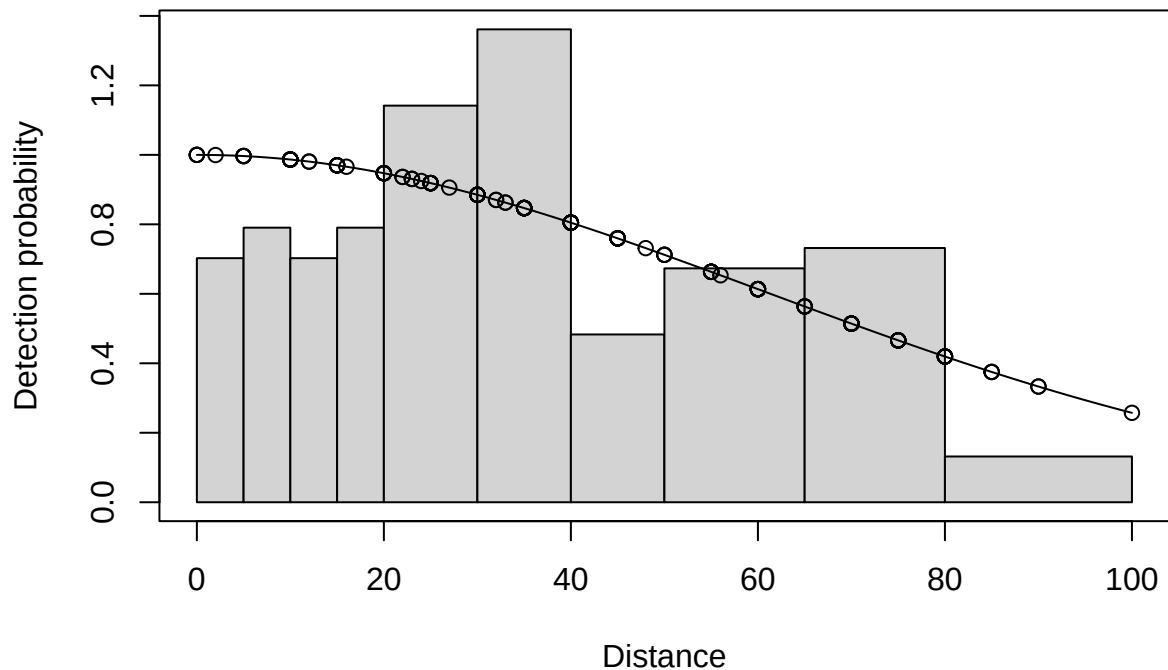
```
## AIC= 1418.188
```

```
summary(wren.hn)
```

```
##
## Summary for distance analysis
## Number of observations : 156
## Distance range       : 0 - 100
##
## Model      : Half-normal key function
## AIC       : 1418.188
## Optimisation: mrds (nlminb)
##
## Detection function parameters
## Scale coefficient(s):
##      estimate      se
## (Intercept) 4.105816 0.1327744
##
##      Estimate      SE      CV
## Average p      0.685037 0.05678821 0.08289802
## N in covered region 227.724931 21.47275208 0.09429250
##
## Summary statistics:
##      Region Area CoveredArea Effort  n k      ER      se.ER      cv.ER
## 1 Montrave 33.2      193.2   9.66 156 19 16.14907 1.226096 0.07592366
##
## Abundance:
##      Label Estimate      se      cv      lcl      ucl      df
## 1 Total 39.13286 4.399007 0.1124121 31.3023 48.9223 74.24595
##
## Density:
##      Label Estimate      se      cv      lcl      ucl      df
## 1 Total 1.1787 0.1325002 0.1124121 0.9428403 1.473563 74.24595
```

```
cutpoints <- c(0,5,10,15,20,30,40,50,65,80,100)
plot(wren.hn, breaks=cutpoints, main="Half normal model, wren line transects")
```


Half normal model, wren line transects



Try other functions

```
wren.unif.cos <- ds(wren_lt, key="unif", adjustment="cos", convert_units=conversion.factor)
```

```
## Starting AIC adjustment term selection.
```

```
## Fitting uniform key function
```

```
## AIC= 1436.813
```

```
## Fitting uniform key function with cosine(1) adjustments
```

```
## AIC= 1421.634
```

```
## Fitting uniform key function with cosine(1,2) adjustments
```

```
## AIC= 1417.792
```

```
## Fitting uniform key function with cosine(1,2,3) adjustments
```

```
## AIC= 1416.431
```

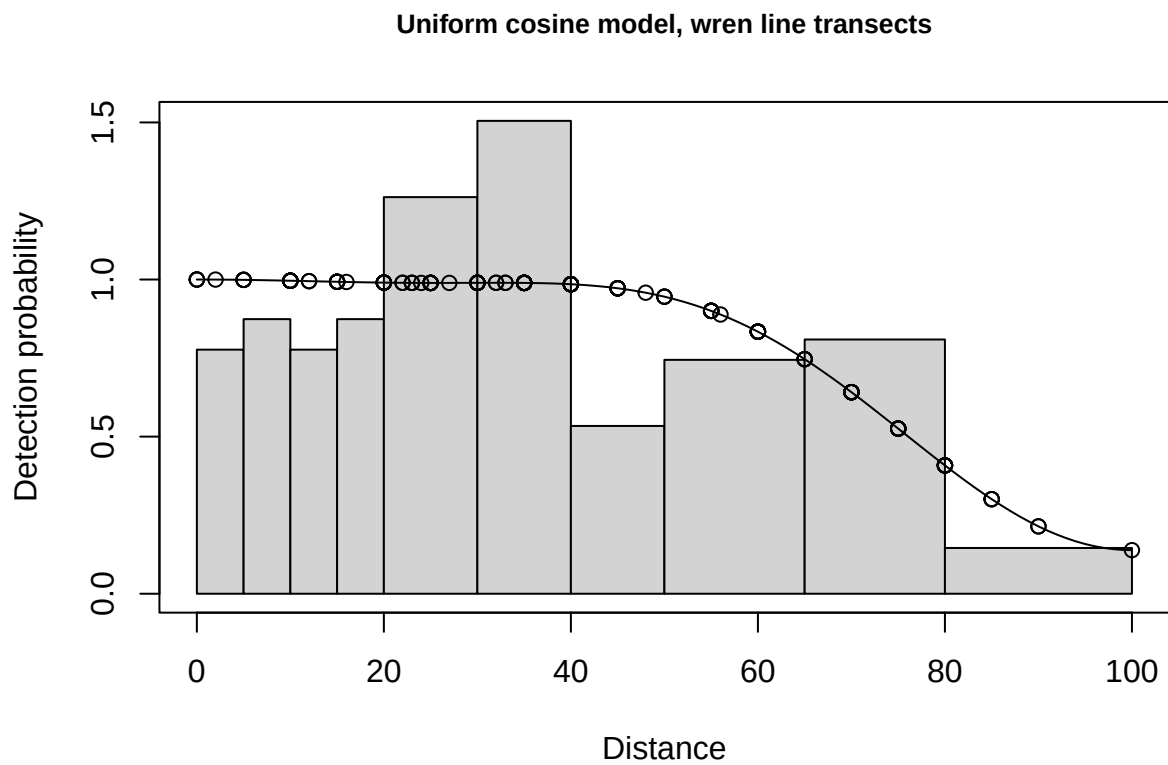
```
## Fitting uniform key function with cosine(1,2,3,4) adjustments
```

```
## AIC= 1416.959
```

```
##
```

```
## Uniform key function with cosine(1,2,3) adjustments selected.
```

```
plot(wren.unif.cos, breaks=cutpoints, main="Uniform cosine model, wren line transects")
```



```
wren.hr.poly <- ds(wren_lt, key="hr", adjustment="poly", convert_units=conversion.factor)
```

```
## Starting AIC adjustment term selection.
```

```
## Fitting hazard-rate key function
```

```
## AIC= 1412.133
```

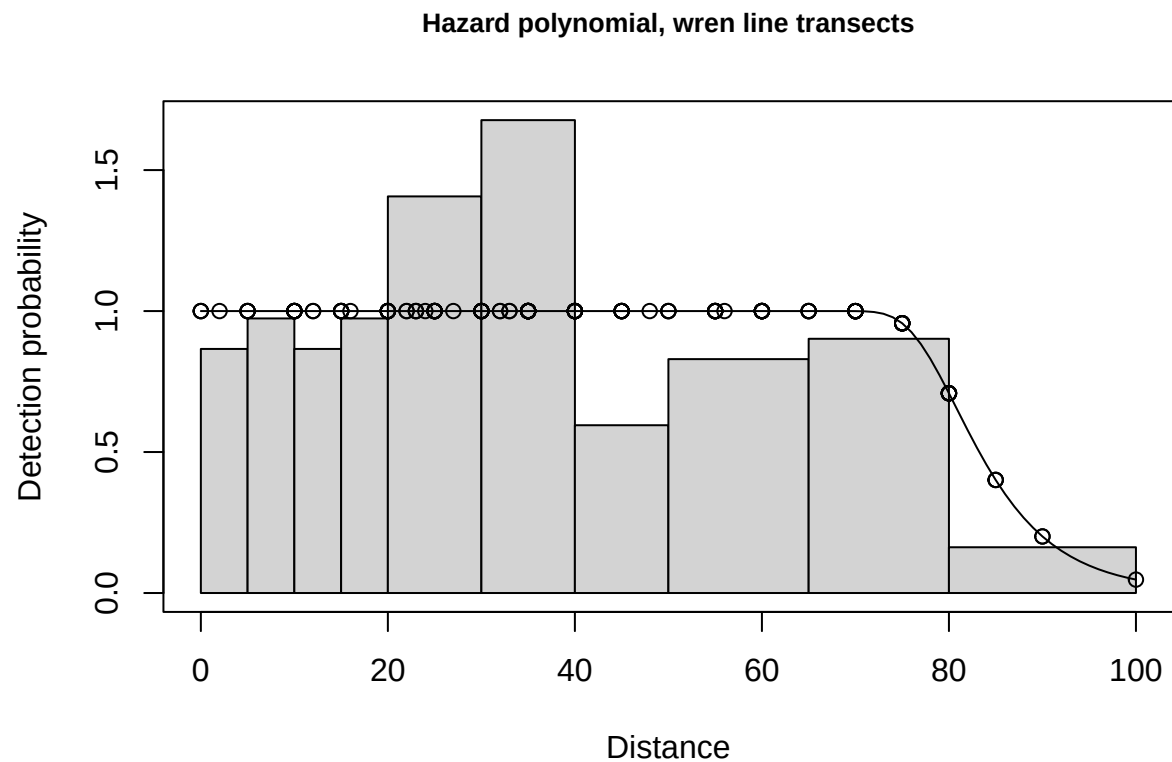
```
## Fitting hazard-rate key function with simple polynomial(4) adjustments
```

```
## AIC= 1413.588
```

```
##
```

```
## Hazard-rate key function selected.
```

```
plot(wren.hr.poly, breaks=cutpoints, main="Hazard polynomial, wren line transects")
```

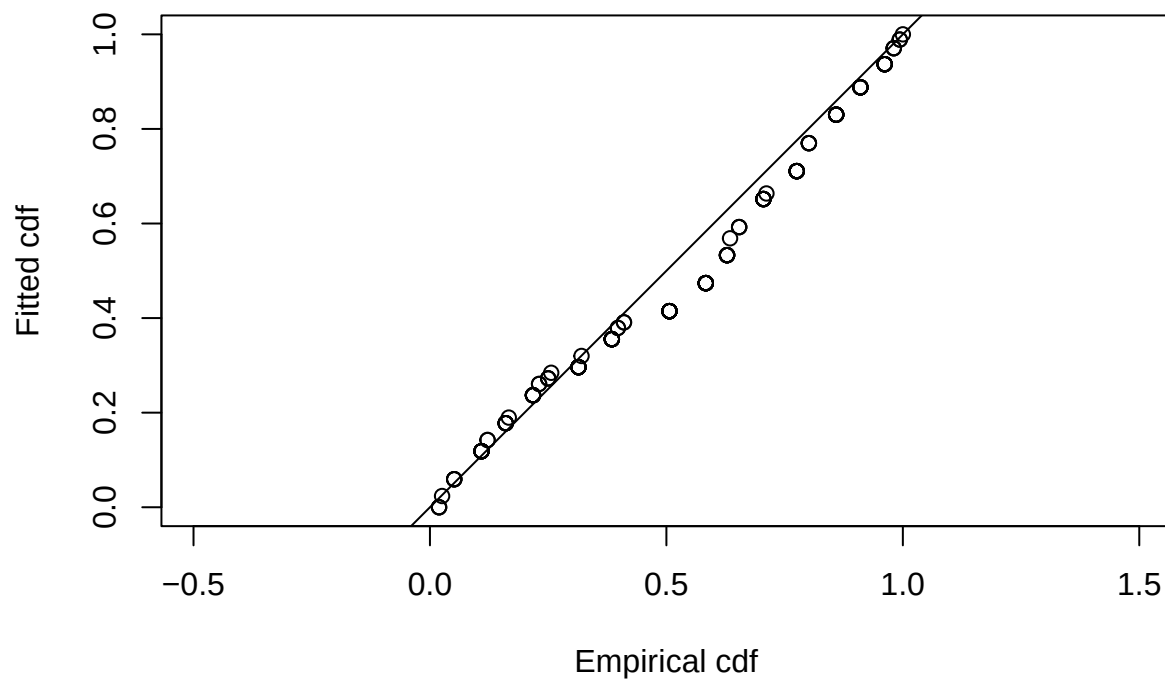


```
AIC(wren.hn, wren.hr.poly, wren.unif.cos)
```

```
##          df      AIC
## wren.hn      1 1418.188
## wren.hr.poly  2 1412.133
## wren.unif.cos  3 1416.431
```

Check fit of best model

```
gof_ds(wren.hr.poly)
```



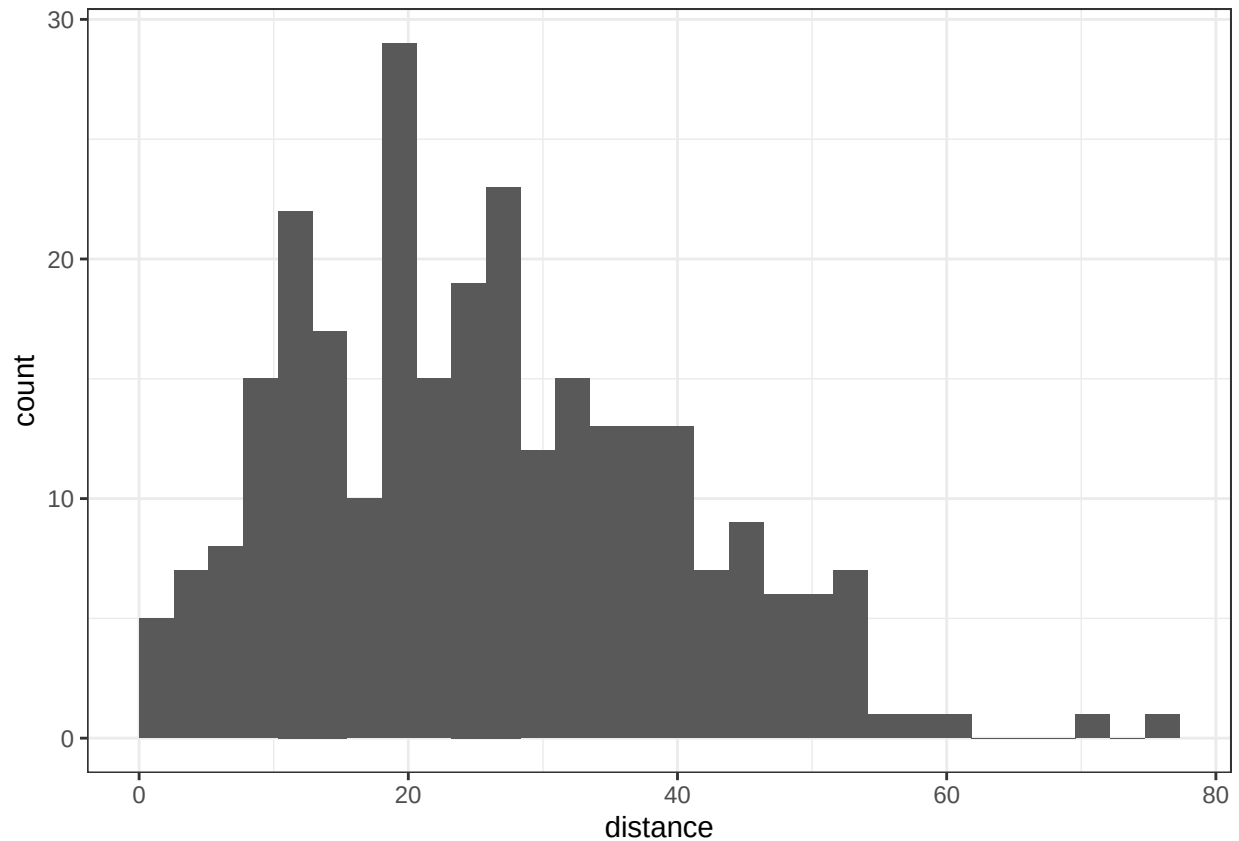
```
##
## Goodness of fit results for ddf object
##
## Distance sampling Cramer-von Mises test (unweighted)
## Test statistic = 0.249897 p-value = 0.188501
```

Point transect with sparrows

```
data(Savannah_sparrow_1980)
Savannah_sparrow_1980$Region.Label <- "Single_stratum"
ggplot(Savannah_sparrow_1980, aes(x=distance)) +
  geom_histogram()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

```
## Warning: Removed 192 rows containing non-finite outside the scale range
## ('stat_bin()').
```



```
##ds
```

Find appropriate detection function

```
sasp.hn <- ds(data=Savannah_sparrow_1980,
  key="hn", #half normal
  adjustment=NULL, #no adjustment based on fitting variables
  transect="point", #data is distance from point of observation.
  #No sampling effort given
  convert_units=convert_units("meter", NULL, "hectare"),
  truncation="5%") #Leave out most distance 5%
```

```
## Fitting half-normal key function
```

```
## AIC= 2021.776
```

```
summary(sasp.hn)
```

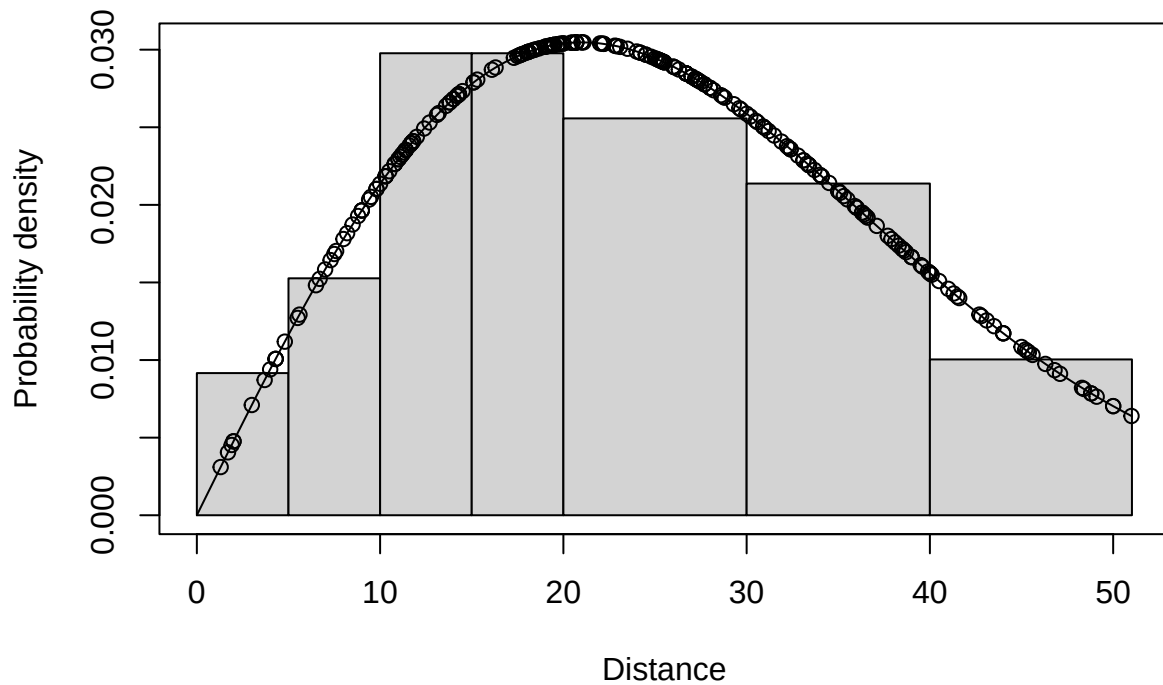
```
##
## Summary for distance analysis
## Number of observations : 262
## Distance range       : 0 - 51.025
##
## Model                : Half-normal key function
```

```
## AIC : 2021.776
## Optimisation: mrds (nlminb)
##
## Detection function parameters
## Scale coefficient(s):
## estimate se
## (Intercept) 3.044624 0.04270318
##
## Estimate SE CV
## Average p 0.321125 0.02296165 0.07150378
## N in covered region 815.881752 71.61153776 0.08777196
##
## Summary statistics:
## Region Area CoveredArea Effort n k ER se.ER
## 1 Single_stratum 1 305.0877 373 262 373 0.7024129 0.04726421
## cv.ER
## 1 0.06728836
##
## Abundance:
## Label Estimate se cv lcl ucl df
## 1 Total 2.674253 0.2625745 0.09818612 2.206266 3.241509 598.5905
##
## Density:
## Label Estimate se cv lcl ucl df
## 1 Total 2.674253 0.2625745 0.09818612 2.206266 3.241509 598.5905
```

```
cutpoints <- c(0,5,10,15,20,30,40,max(Savannah_sparrow_1980$distance, na.rm=TRUE))
plot(sasp.hn, breaks=cutpoints, pdf=TRUE, main="Savannah sparrow point transect data.")
```

```
## Specified endpoints > 51.024999625; values reset.
```

Savannah sparrow point transect data.



With a different detection function

```
sasp.unif.cos <- ds(Savannah_sparrow_1980, key="unif", adjustment="cos",
  transect="point",
  convert_units=convert_units("meter", NULL, "hectare"), truncation="5%")
```

```
## Starting AIC adjustment term selection.
```

```
## Fitting uniform key function
```

```
## AIC= 2176.298
```

```
## Fitting uniform key function with cosine(1) adjustments
```

```
## AIC= 2023.178
```

```
## Fitting uniform key function with cosine(1,2) adjustments
```

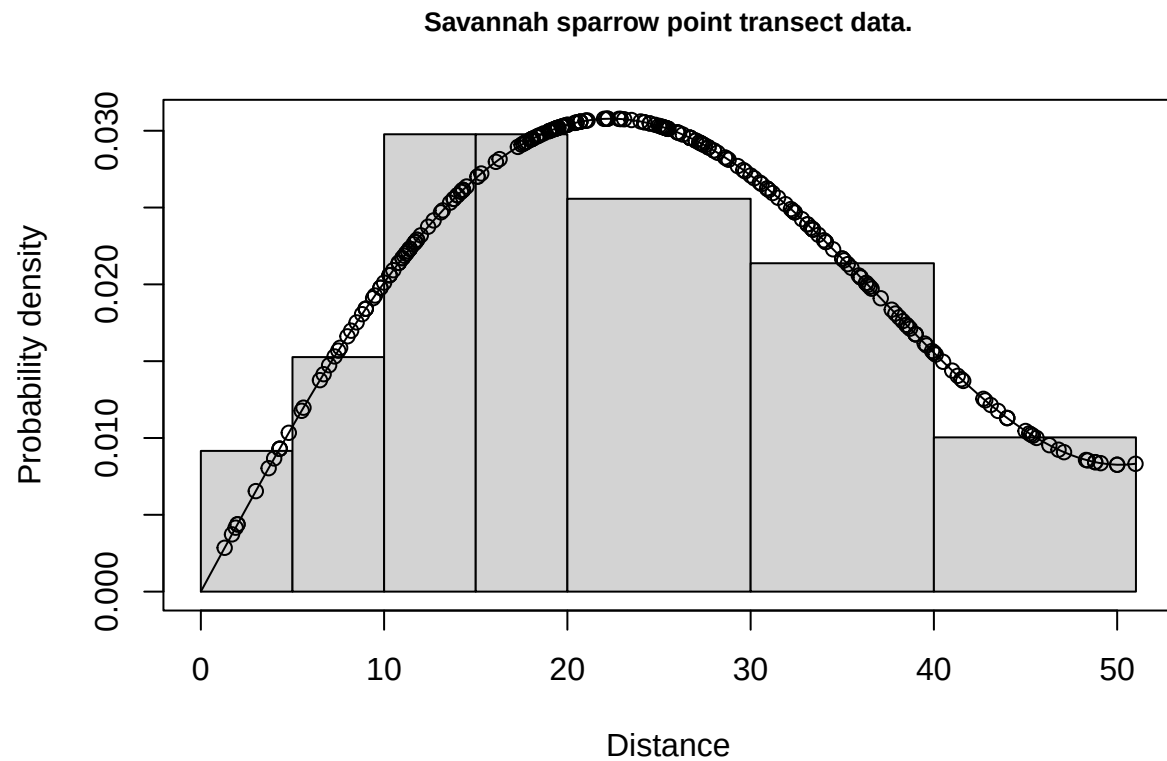
```
## AIC= 2023.745
```

```
##
```

```
## Uniform key function with cosine(1) adjustments selected.
```

```
plot(sasp.unif.cos, breaks=cutpoints, pdf=TRUE, main="Savannah sparrow point transect data.")
```

```
## Specified endpoints > 51.024999625; values reset.
```



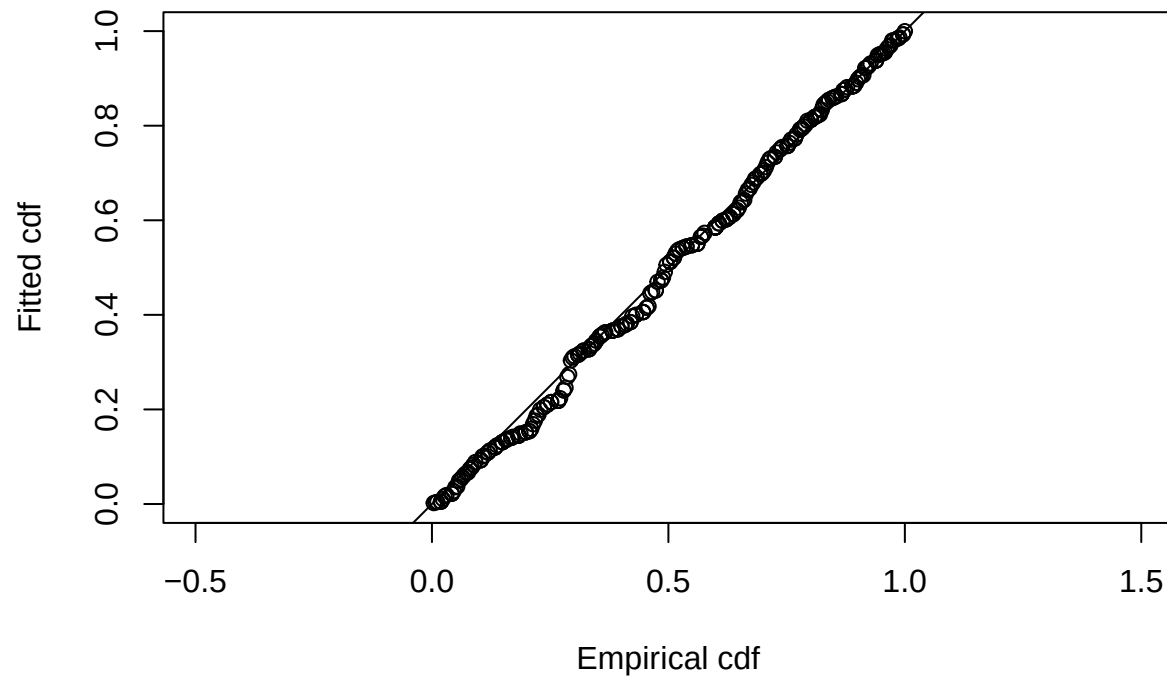
Compare models

```
AIC(sasp.hn, sasp.unif.cos)
```

```
##           df      AIC
## sasp.hn      1 2021.776
## sasp.unif.cos 1 2023.178
```

Half normal is best


```
gof_ds(sasp.hn)
```



```
##  
## Goodness of fit results for ddf object  
##  
## Distance sampling Cramer-von Mises test (unweighted)  
## Test statistic = 0.0835959 p-value = 0.671325
```

Residuals look fine